An Exploratory Analysis of Participation Patterns in European Citizens' Initiatives

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INTRODUCTION

The European Citizens' Initiative (ECI) allows citizens to become more involved in EU democratic processes and policy development. Each ECI is essentially a petition that can propose changes to the EU Commission. To be considered by the European Commission, an initiative must collect at least one million signatures and achieve minimum thresholds across at least seven member states.

Participation rates vary greatly between initiatives and countries, this project aims to analyse and explain disparities in participation rates across initiatives and countries. This phase of the project seeks to investigate whether country level socio-economic factors can explain differences in participation rates.

RESEARCH QUESTIONS

1. How do participation levels differ across European Citizens' Initiatives?
2. To what extent do country level socio-economic factors explain variation in ECI participation?
3. Are differences in participation more strongly associated with petition specific characteristics than country level socio-economic factors?

PRELIMINARY HYPOTHESES

Variance between Petitions
- H_0: There are no significant differences in participation rates between petitions.
- H_1: Participation rates differ significantly between petitions.

Raw Signature Counts
- H_0: Population size and economic indicators do not explain a significant proportion of the variation in raw signature counts.
- H_1: Population size and economic indicators explain a significant portion of the variation in raw signature counts.

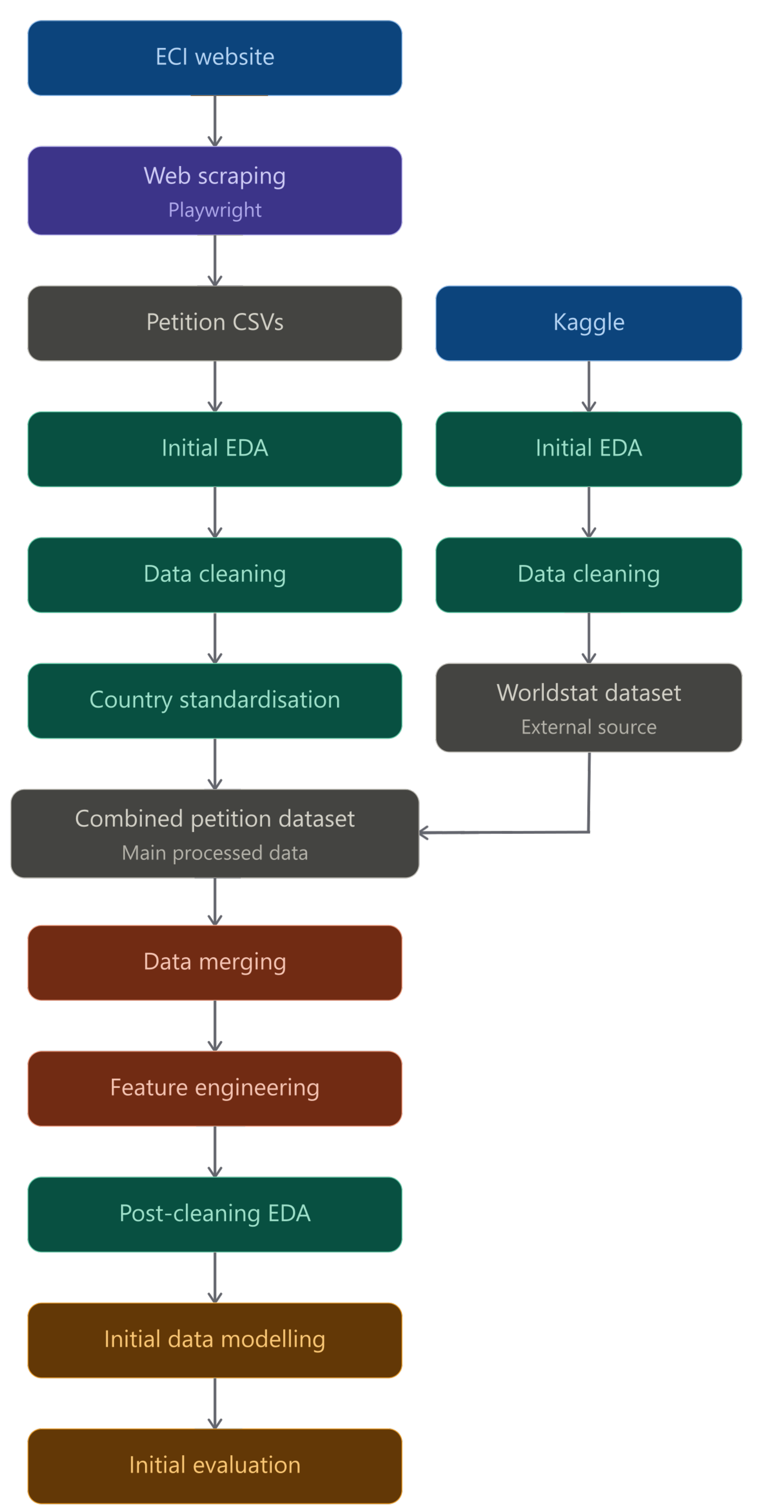
Signatures Per Capita
- H_0: Socio-economic indicators have no significant relationship with participation rates when participation is normalised by population.
- H_1: Socio-economic indicators have limited explanatory power when participation is normalised by population.

LITERATURE REVIEW

- Successful ECIs often rely on mobilisation within larger member states (Weisskircher, 2019).
- UK Petition participation differs significantly between campaigns (Hale et al., 2013).
- Socio-economic factors alone provide limited explanatory power (Giebler et al., 2024).

Research Gap
Limited research combines ECI petition data with socio-economic variables to analyse participation patterns using statistical modelling techniques.

METHODOLOGY



Methodology adopted CRISP-DM

DATA COLLECTION

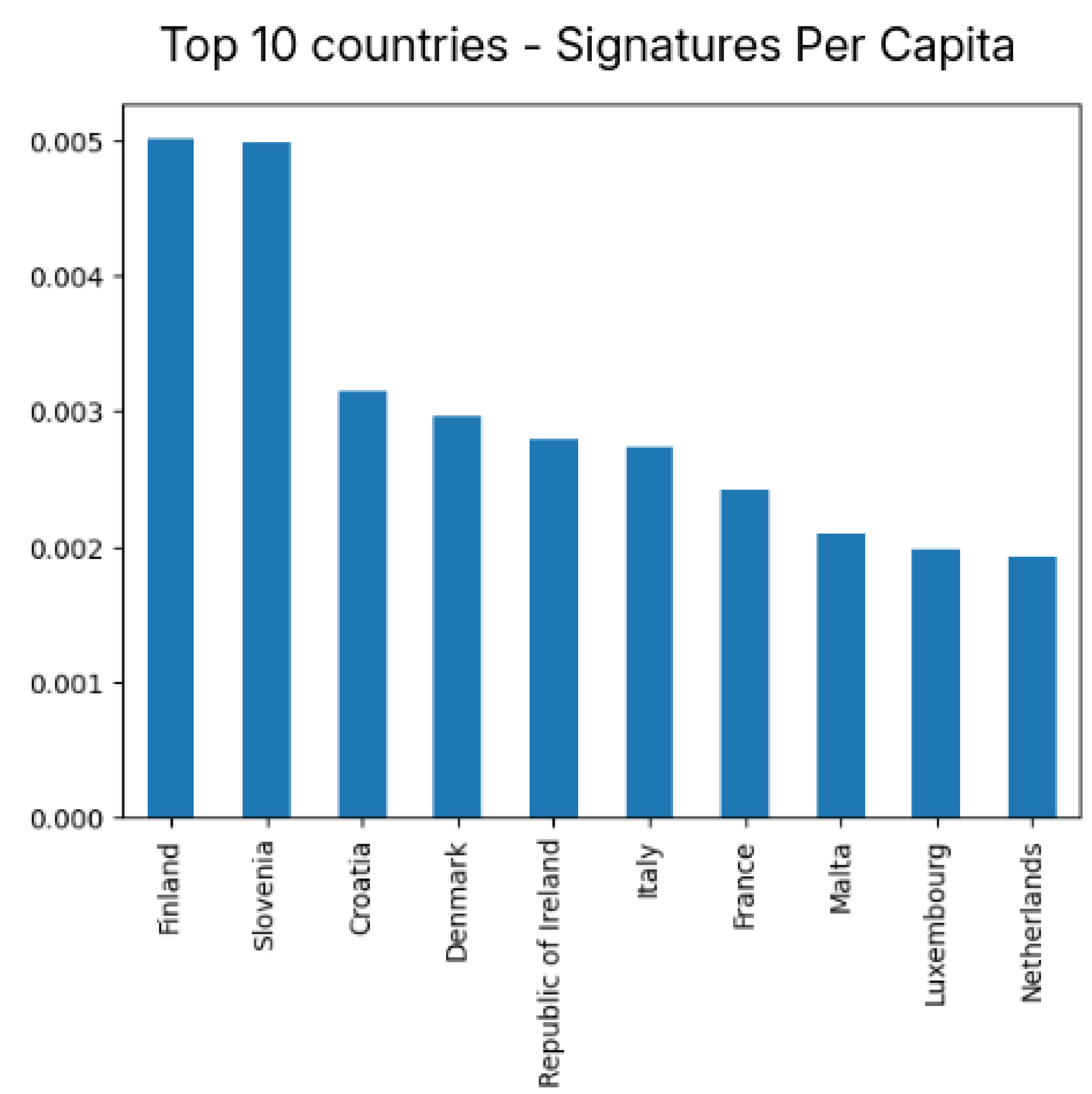
European Citizens' Initiative Data

Seven initiatives were collected using a custom Python scraper built with Playwright.
- Ban on Conversion Practices
- Defence of Agriculture and Rural Economy
- My Voice My Choice
- Air Quotas
- Stop Destroying Videogames
- Stop Cruelty Stop Slaughter
- Stop Fake Food: Origin on Label

External Dataset

World Data 2023 dataset obtained from Kaggle.

INITIAL EXPLORATORY DATA ANALYSIS



SELECTED VARIABLES

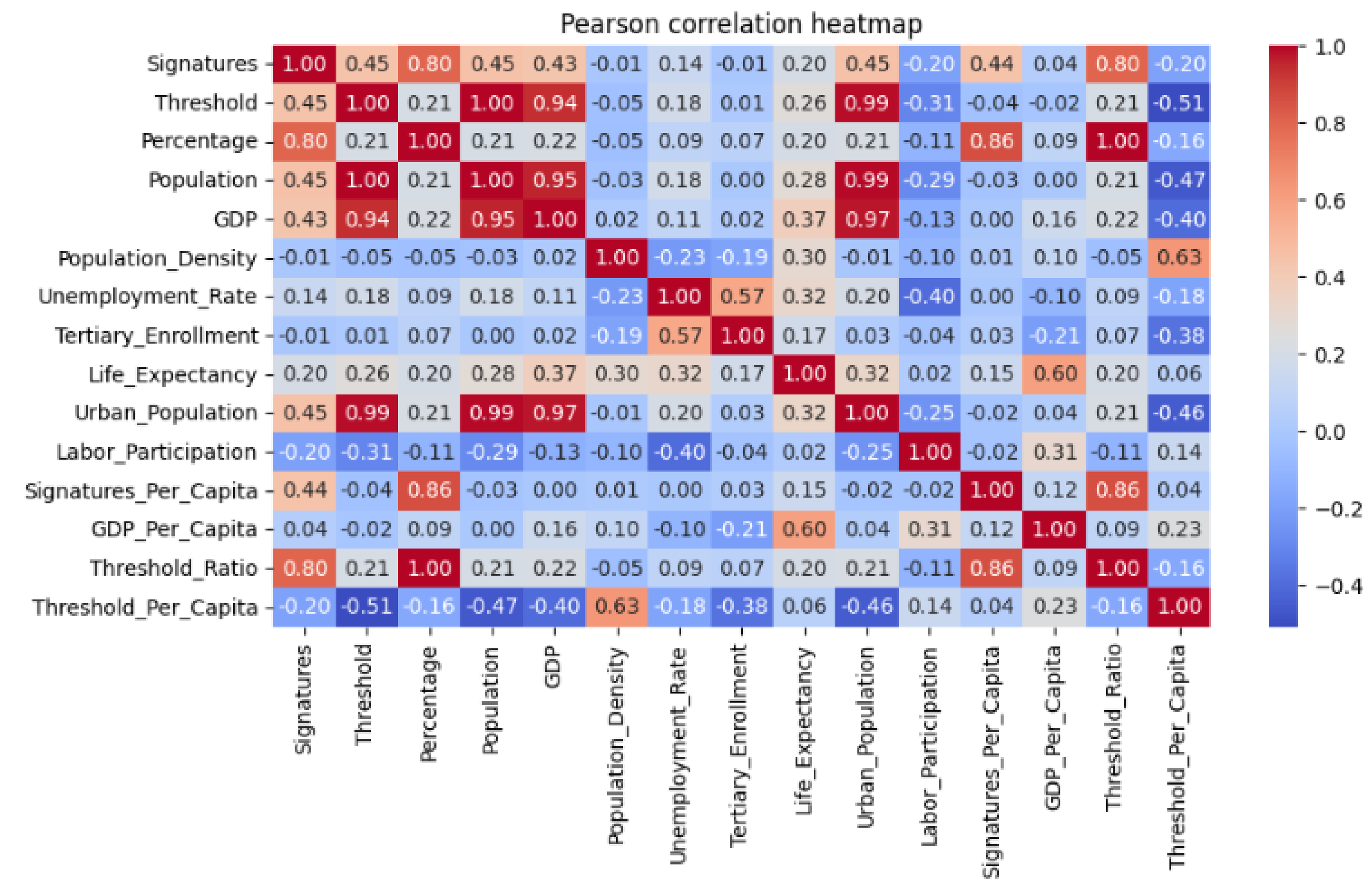
- Country Name
- Number of Signatures
- Country Minimum Threshold
- Percentage of Threshold Reached
- Population
- GDP
- Unemployment Rate
- Life Expectancy
- Population Density
- Gross Tertiary Education Enrollment
- Urban Population
- Labor Participation

DATA PREPARATION

- Merging of ECI datasets by creating a Petition ID
- Checking and Handling for Missing Values and Outliers
- Removal of Non Country specifc rows from ECI datasets
- Standardisation of country names
- Conversion of numeric fields
- Merging with Worldstat Data using Country Name
- Feature engineering

Feature Engineering
- Signatures Per Capita = Signatures / Population
- GDP Per Capita = GDP / Population
- Threshold Ratio = Signatures / Threshold
- Threshold Per Capita = Threshold / Population

POST CLEANING EXPLORATORY DATA ANALYSIS



Key Findings
- Population moderately correlates with total signatures.
- GDP moderately correlates with total signatures.
- Many socio-economic variables weakly correlate with signatures per capita.
- Participation per capita varies greatly between initiatives.

INITIAL DATA MODELLING

ANOVA Test

$$p < 0.05$$

Participation rates differ significantly between petitions.

Linear Regression Model 1

$$\text{Signatures} \sim \text{Population} + \text{GDP}$$

$$R^2 = 0.256$$

Population and GDP explain roughly 25.6% of variation in total signature counts.

Linear Regression Model 2

$$\text{Signatures Per Capita} \sim \text{GDP Per Capita} + \text{Unemployment} + \text{Life Expectancy} + \text{Population Density} + \text{Tertiary Enrollmen}$$

$$R^2 = -0.045$$

Socio-economic factors alone provide little explanatory power.

KEY FINDINGS

The modelling results indicated that petition specific factors may be more important than country level socio-economic factors when explaining participation rates. This conclusion is consistent with the findings reported by Weisskircher (2019) and Giebler et al. (2025) who both emphasised the importance of campaign context and issue specific factors.

FUTURE WORK

- Increase the number of analysed initiatives.
- Principal Component Analysis (PCA).
- Clustering of member states.
- Additional machine learning models.
- Model comparison and evaluation.
- Flask dashboard for visualisation.

REFERENCES

Weisskircher, M. (2019). The european citizens' initiative: Mobilization strategies and consequences. https://doi.org/10.1177/0032321719859792.

Hale, S. A., Margetts, H., & Yasseri, T. (2013). Petition growth and success rates on the uk no. 10 downing street website. https://doi.org/10.1145/2464464.2464518.

Giebler, H., Giesecke, J., Humphreys, M., Hutter, S., & Klüver, H. (2025). Mobilizing europe's citizens to take action on migration and climate change: behavioral evidence from 27 eu member states. https://doi.org/10.1080/13501763.2025.2512032.

Lnenicka, M., Nikiforova, A., Luterek, M., Milic, P., Rudmark, D., Neumaier, S., . . . Pedro, M. (2023). Identifying patterns and recommendations of and for sustain- able open data initiatives: a benchmarking-driven analysis of open government data initiatives among european countries. https://arxiv.org/abs/2312.00551.

Geißel, B., Krämling, A., & Paulus, L. (2023). Direct democracy and equality: context is the key. https://doi.org/10.1057/s41269-023-00316-4.